








# Statistics of Financial Markets

## Chapter 3: $\alpha$-stable distributions




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# Acronyms used in this material

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **BET**: Bucharest Exchange Trading (index)
- **CLT**: Central Limit Theorem
- **CMS**: Chambers–Mallows–Stuck (simulation method)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **FFT**: Fast Fourier Transform
- **GCLT**: Generalised Central Limit Theorem
- **IBM**: International Business Machines (company)
- **KS**: Kolmogorov–Smirnov (test)
- **ML**: Maximum Likelihood
- **QQ**: Quantile–Quantile (plot)
- **SAS**: Statistical Analysis System (software)
- **SE**: Standard Error

# Today's Question, Route and Learning Outcomes

- **Question**: which probability laws can describe daily returns and also their weekly and monthly sums?
  - Chapter 2: daily returns have heavier tails than the Normal distribution
  - log returns add up over time (Chapter 1): the law of a sum matters
- **Route and outcomes**: after this chapter you can
  - state the stability property and compute the scale of a sum of stable variables
  - explain the generalised central limit theorem and when sums do not approach the Normal distribution
  - read the four parameters $(\alpha, \beta, \gamma, \delta)$ and convert between the S0 and S1 parameterisations
  - say which moments exist for a given $\alpha$ and compare tail probabilities with the Normal distribution
  - simulate stable variables and estimate their parameters by quantiles and by maximum likelihood
  - judge, on BET, S&P 500, DAX and Bitcoin returns, where the stable model works and where it fails

## Reading and Tools

- Main reference: Nolan (2020), Ch. 1 (definitions, parameterisations, tails) and Ch. 4 (estimation)
  - short survey with code: Borak, Härdle & Weron (2005); updated in Borak, Misiorek & Weron (2011)
- Textbook: Franke, Härdle & Hafner (2019), Ch. 3 (probability) and Ch. 18 (heavy tails and extreme risks)
- The original papers: Mandelbrot (1963), Fama (1965)
- Python Quantlets of this chapter: Quantlets/Ch_03
  - ported from the Quantlet sim_stable; each chart links to the Quantlet that draws it
- Lecture notebook: open in Google Colab
- Video course: Stable Distribution

# Cotton: an Old Market with Long Price Records

- In the 19th century cotton was one of the most traded commodities in the world
  - exchanges in New Orleans, New York and Liverpool quoted prices every day
  - the result: some of the longest daily and monthly price series in economics
- In the early 1960s these series reached Benoit Mandelbrot, then at IBM
  - he looked at the **changes** of the logarithm of the price, i.e. log returns
- His paper Mandelbrot (1963) started the study of heavy tails in finance



Edgar Degas, A Cotton Office in New Orleans (1873)

Image: Edgar Degas (1873); public domain; Wikimedia Commons

## What Mandelbrot Saw

- **Too many large changes**: far more big moves than the Normal distribution allows
  - the sample variance did not settle as the sample grew
- **The same shape at every horizon**: daily and monthly changes looked alike after rescaling
  - a sum of many daily changes kept the shape of one daily change
- **His proposal**: stable Paretian laws with $\alpha \approx 1.7$ Mandelbrot (1963)
  - "Paretian": the tails decay like a power of $x$, as in Pareto's law of incomes
  - consequence: the variance of price changes would be infinite



Benoit Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 fr; Wikimedia Commons

## Fama Tests the Idea on Stocks

- Fama (1965): daily returns of the 30 stocks of the Dow Jones Industrial Average
  - more large returns than the Normal distribution predicts, for every stock
  - estimates of $\alpha$ below 2: support for Mandelbrot's hypothesis
- Estimation methods for symmetric stable laws followed
  - Fama & Roll (1968), Fama & Roll (1971): quantile-based estimates and tables
- Within ten years the evidence started to turn: see the critique at the end of the chapter



Eugene Fama at the Nobel Prize ceremony, 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

## Why the Law of a Sum Matters

- A monthly log return is the sum of about 21 daily log returns
  - a model for daily returns implies a model for weekly, monthly and yearly returns
- The Normal distribution keeps its shape under summation (Chapter 2)
  - but it has thin tails: it misses the large daily moves
- **Question for the room**: the Student-t distribution has heavy tails; is a sum of 21 independent Student-t variables again Student-t?
- **Answer**
  - no: the Student-t family is not closed under summation
  - if its variance is finite, the sum moves towards the Normal distribution (the central limit theorem)
- **Goal**: find every law whose sums keep the same shape

## Definition: Stable Laws

- $X$ is **stable** if, for $X_1, X_2$ independent copies of $X$ and any $a, b > 0$, there are $c > 0$ and $d$ with
  - $aX_1 + bX_2 \stackrel{d}{=} cX + d$, where $\stackrel{d}{=}$ means "has the same distribution as"
  - $a, b$: the weights of the two copies; $c$: the scale and $d$: the shift of the result
  - **strictly stable** if $d = 0$ for all $a, b$
- In words: a weighted sum of two copies has the same shape as one copy
  - only the scale ($c$) and the location ($d$) change
- By induction, for $n$ independent copies: $X_1 + \cdots + X_n \stackrel{d}{=} c_n X + d_n$
  - the only possible scale factor is $c_n = n^{1/\alpha}$ with $0 < \alpha \leq 2$ Nolan (2020)
- $\alpha$ is called the **index of stability** (also: characteristic exponent, tail index)

## Two Examples: Normal and Cauchy

- **Normal**: $X_1, X_2 \sim N(0, \sigma^2)$ independent
  - $aX_1 + bX_2 \sim N(0, (a^2 + b^2)\sigma^2)$, so $c = \sqrt{a^2 + b^2}$
  - variances add: $c^2 = a^2 + b^2$, the case $\alpha = 2$
- **Cauchy**: density $f(x) = \dfrac{1}{\pi(1 + x^2)}$
  - $aX_1 + bX_2$ is Cauchy with scale $c = a + b$
  - scales add linearly: $c^1 = a^1 + b^1$, the case $\alpha = 1$
  - the average of $n$ Cauchy draws is again standard Cauchy: averaging does not help



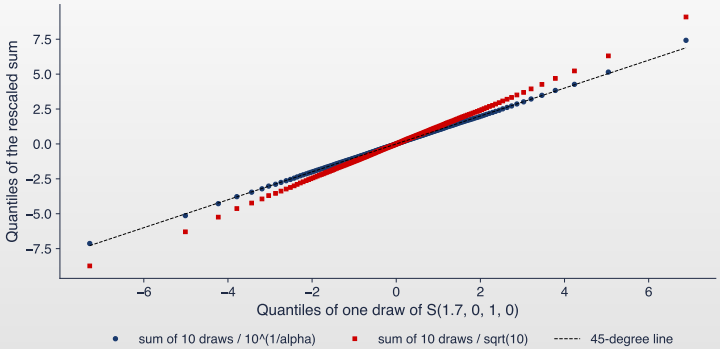

Augustin-Louis Cauchy (1789–1857)

Lithograph: J. Boilly (1821), Wellcome Collection; CC BY 2.0; Wikimedia Commons

## The General Rule and a Worked Example

- For $X_1, X_2$ independent copies of a strictly stable $X$ with index $\alpha$:
  - $c^\alpha = a^\alpha + b^\alpha$, i.e. $c = (a^\alpha + b^\alpha)^{1/\alpha}$
- **Example**: $a = 3$, $b = 4$
  - $\alpha = 2$: $c = \sqrt{9 + 16} = 5.00$
  - $\alpha = 1.5$: $c = (3^{1.5} + 4^{1.5})^{1/1.5} = 5.58$
  - $\alpha = 1$: $c = 3 + 4 = 7.00$
- The smaller $\alpha$, the larger the scale of the sum: one large term can dominate the sum
- **What do you think?** For $\alpha = 1.5$, what is the scale factor of a sum of two copies?
- **Answer**
  - $c_2 = 2^{1/1.5} = 2^{2/3} = 1.59$, larger than $\sqrt{2} \approx 1.41$ for the Normal distribution

## Stability in a Simulation




- Notation (Section 4): $S(\alpha, \beta, \gamma, \delta)$ is the stable law with index $\alpha$, skewness $\beta$, scale $\gamma$ and location $\delta$
- Blue: sums of 10 draws of $S(1.7, 0, 1, 0)$ divided by $10^{1/1.7} = 3.87$ lie on the 45-degree line: same law as one draw
- Red: dividing by $\sqrt{10} = 3.16$ (the Normal rule) leaves the tails too wide: 99% quantile 6.31 vs 5.04

SFM_ch3_stability_gclt

## Consequences: Horizons and Diversification

- **Horizon**: if daily returns are i.i.d. (independent, identically distributed) $S(\alpha, 0, \gamma, 0)$, the 20-day sum has scale $20^{1/\alpha}\gamma$
  - $\alpha = 2$: $4.47\,\gamma$ (the square-root-of-time rule); $\alpha = 1.7$: $5.83\,\gamma$; $\alpha = 1.5$: $7.37\,\gamma$
  - with $\alpha < 2$, the risk of long horizons grows faster than $\sqrt{n}$
- **Diversification**: an equally weighted portfolio of $N$ i.i.d. stable assets has scale $N^{1/\alpha - 1}\gamma$
  - $N = 10$: $\alpha = 2$ gives $0.32\,\gamma$; $\alpha = 1.5$ gives $0.46\,\gamma$; $\alpha = 1$ gives $1.00\,\gamma$
  - with $\alpha = 1$ diversification does not reduce the scale at all; Fama (1965) discussed this point

## Recap: Stability

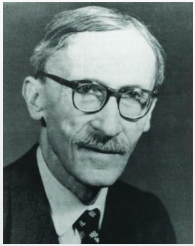
- Stable law: a sum of independent copies has the same shape, only rescaled and shifted
- Scale rule: $c^\alpha = a^\alpha + b^\alpha$; $n$ copies: factor $n^{1/\alpha}$, $0 < \alpha \leq 2$
- Normal: $\alpha = 2$; Cauchy: $\alpha = 1$
- Small $\alpha$: risk grows faster with the horizon and diversification helps less

## From the Classical to the Generalised CLT

- **Classical CLT** (central limit theorem, Chapter 2): $X_i$ i.i.d. with mean $\mu$ and finite variance $\sigma^2$
  - $\dfrac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \to N(0,1)$ in distribution
- What if the variance is infinite? The normalisation $\sqrt{n}$ is then wrong
- **GCLT** (generalised central limit theorem) Gnedenko & Kolmogorov (1954): if $(X_1 + \cdots + X_n - b_n)/a_n$ converges to a non-degenerate law, that law is stable
  - $a_n > 0$, $b_n$: normalising constants that rescale and recentre the sum (for the classical CLT, $a_n = \sigma\sqrt{n}$, $b_n = n\mu$)
  - stable laws are the **only** possible limits of normalised sums of i.i.d. variables
  - the Normal distribution is the special case $\alpha = 2$

## Domains of Attraction

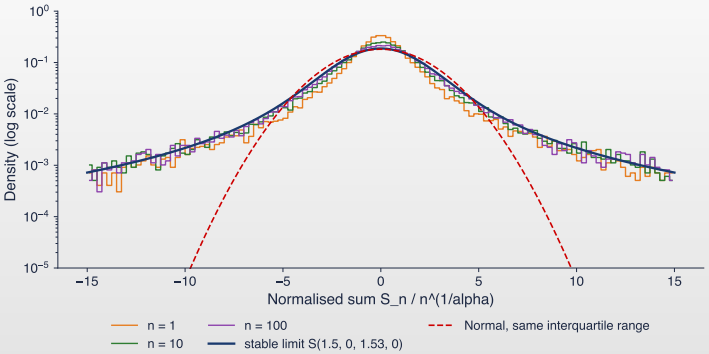
- The **domain of attraction** of a stable law: all laws whose normalised sums converge to it
  - finite variance: domain of attraction of the Normal distribution, $a_n \propto \sqrt{n}$
  - power-law tails $P(|X| > x) \approx Cx^{-\alpha}$ with $0 < \alpha < 2$: the stable law with the same $\alpha$, $a_n \propto n^{1/\alpha}$
  - $C > 0$: a constant; $\propto$: proportional to
- So the **tail** of one return decides the law of long sums
- Paul Lévy characterised the stable laws in the 1920s; the full theory is in Gnedenko & Kolmogorov (1954)



Paul Lévy (1886–1971)

Photo: Konrad Jacobs, Oberwolfach Photo Collection; CC BY-SA 2.0 de; Wikimedia Commons

# The GCLT in a Simulation




- Student-t with $\nu = 1.5$ degrees of freedom: infinite variance, tails $\propto x^{-1.5}$; sums of $n$ draws divided by $n^{1/1.5}$
- $n = 1, 10, 100$: the histograms approach the stable limit $S(1.5, 0, 1.53, 0)$; $P(|\cdot| > 10)$: 2.73% for $n = 100$, 2.64% in the limit
- A Normal law with the same interquartile range gives 0.0005%: it misses the tails completely

SFM_ch3_stability_gclt

## What the GCLT Means for Returns

- Two possible worlds for daily returns
  - **tail index below 2** (infinite variance): monthly returns are stable with the **same** $\alpha$
  - **tail index above 2** (finite variance): monthly returns move towards the Normal distribution
- The two worlds give different testable predictions about returns at longer horizons
- **Question for the room**: in which world do the S&P 500 returns live?
- **Answer**
  - we test it with real data in the last sections: estimates of $\alpha$ at daily, weekly and monthly horizons

## Recap: The Generalised CLT

- Classical CLT: finite variance, normalisation $\sqrt{n}$, Normal limit
- GCLT: stable laws are the only limits of normalised sums
- Power-law tails with index $\alpha < 2$: normalisation $n^{1/\alpha}$, stable limit with the same $\alpha$
- The tail of a single return decides the law of long sums

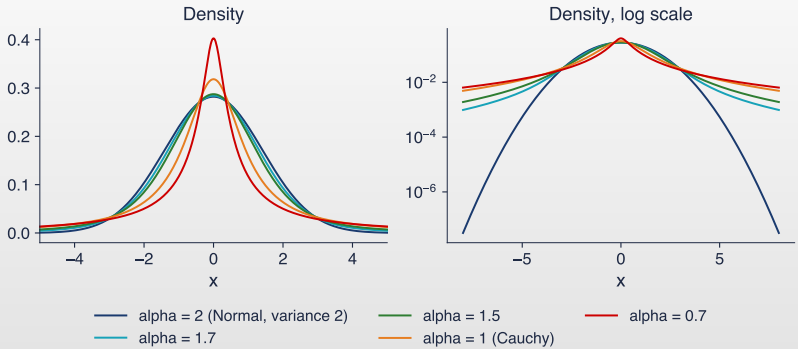
## The Characteristic Function

- **Characteristic function** of $X$: $\varphi_X(t) = E[e^{itX}] = E[\cos(tX)] + i\,E[\sin(tX)]$, $t \in \mathbb{R}$
  - $i = \sqrt{-1}$: the imaginary unit; $t$: a real argument, not time
  - it always exists, because $|e^{itX}| = 1$, even when no moment exists
  - it determines the distribution uniquely
- Two properties used in this chapter
  - independent $X, Y$: $\varphi_{X+Y}(t) = \varphi_X(t)\,\varphi_Y(t)$: sums become products
  - $\varphi_{aX+b}(t) = e^{ibt}\varphi_X(at)$
- Normal $N(\mu, \sigma^2)$: $\varphi(t) = \exp(i\mu t - \sigma^2 t^2/2)$
- Why we need it: apart from three cases, stable laws have **no closed-form density**; they are defined by $\varphi$

## The Stable Characteristic Function (S1)

- $X \sim S(\alpha, \beta, \gamma, \delta; 1)$ if, for $\alpha \neq 1$, Samorodnitsky & Taqqu (1994), Nolan (2020):
  - $\varphi(t) = \exp\Big\{ -\gamma^\alpha |t|^\alpha \big[1 - i\beta \operatorname{sign}(t) \tan \tfrac{\pi\alpha}{2}\big] + i\delta t \Big\}$
  - for $\alpha = 1$: $\varphi(t) = \exp\Big\{ -\gamma|t|\big[1 + i\beta \tfrac{2}{\pi} \operatorname{sign}(t) \ln|t|\big] + i\delta t \Big\}$
- The four parameters
  - $\alpha \in (0, 2]$: **index of stability**; smaller $\alpha$, heavier tails
  - $\beta \in [-1, 1]$: **skewness**; $\beta > 0$ heavier right tail, $\beta < 0$ heavier left tail
  - $\gamma > 0$: **scale** (it plays the role of $\sigma$); $\delta \in \mathbb{R}$: **location**
- $\operatorname{sign}(t)$: $+1$ for $t > 0$, $-1$ for $t < 0$; the term with $\beta$ makes the law asymmetric
- Stability is visible: $\varphi(t)^n$ has the same form, with $\gamma^\alpha$ replaced by $n\gamma^\alpha$

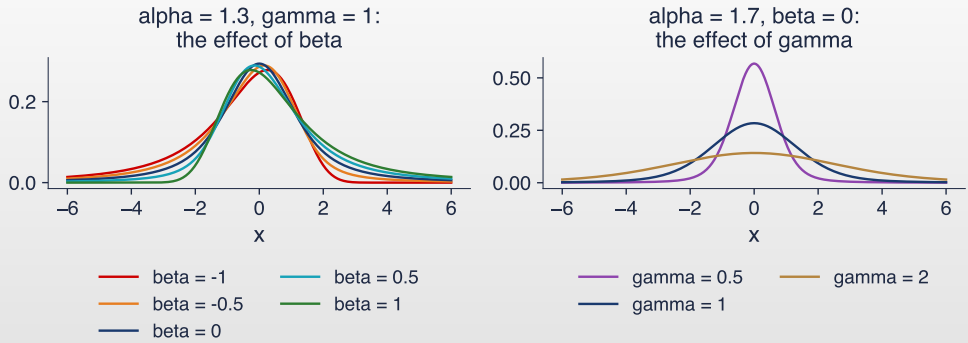
# The Effect of $\alpha$




- Symmetric laws $S(\alpha, 0, 1, 0)$: smaller $\alpha$ gives a taller, narrower peak and much heavier tails
- Log scale: the Normal tail ($\alpha = 2$) falls like $e^{-x^2/4}$; the others fall like a power of $x$

SFM_ch3_densities

# The Effect of $\beta$ and $\gamma$



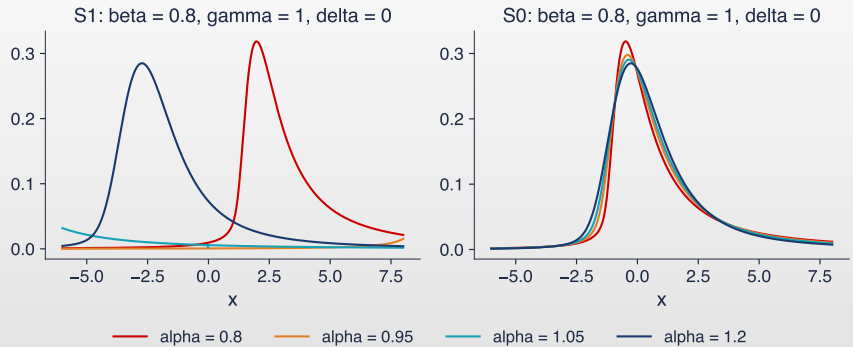

- $\beta$ moves weight between the tails: $\beta = 1$ (green) has the heavier right tail, $\beta = -1$ (red) the heavier left tail
- $\gamma$ stretches the density: doubling $\gamma$ doubles every quantile around $\delta$

SFM_ch3_densities

## Two Parameterisations: S0 and S1

- **S0** (Nolan's recommendation for data analysis), $\alpha \neq 1$:
  - $\varphi(t) = \exp\Big\{ -\gamma^\alpha |t|^\alpha \big[1 + i\beta \operatorname{sign}(t) \tan\tfrac{\pi\alpha}{2} \big(|\gamma t|^{1-\alpha} - 1\big)\big] + i\delta_0 t \Big\}$
- Only the location differs: $\delta_0 = \delta_1 + \beta\gamma \tan\tfrac{\pi\alpha}{2}$ $(\alpha \neq 1)$
  - $\delta_0$, $\delta_1$: the location parameter in S0 and in S1; $\alpha$, $\beta$ and $\gamma$ are the same in both
  - for $\beta = 0$ the two coincide
- Why two?
  - S1: simpler algebra for sums; used in most theory books Samorodnitsky & Taqqu (1994)
  - S0: the density changes continuously with $\alpha$ and $\beta$; better for estimation and plots Nolan (2020)

# S1 vs S0 near $\alpha = 1$




- S1 (left): with $\beta = 0.8$, the density runs away to $+\infty$ as $\alpha \uparrow 1$ and comes back from $-\infty$ as $\alpha \downarrow 1$, because $\tan \frac{\pi\alpha}{2}$ explodes
- S0 (right): the same four laws, shifted; the mode stays near 0 and changes smoothly

SFM_ch3_densities

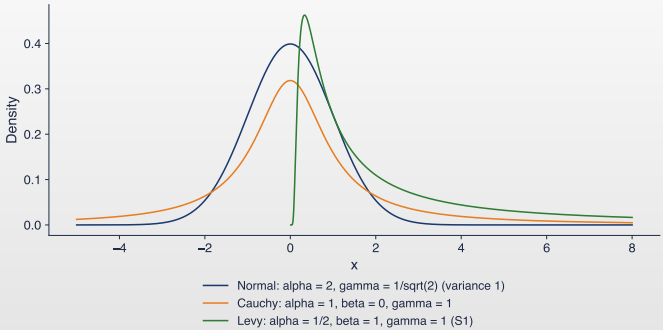
## Mind the Parameterisation in Software

- `scipy.stats.levy_stable` uses **S1** by default
  - switch with `levy_stable.parameterization = 'S0'`
  - arguments: `levy_stable.pdf(x, alpha, beta, loc=delta, scale=gamma)`
- Papers and programs report $\delta$ in S0, in S1 or in other parameterisations: check which one before comparing
- Scale trap: $S(2, 0, \gamma, \delta)$ is $N(\delta, 2\gamma^2)$, not $N(\delta, \gamma^2)$
  - so $\gamma = \sigma/\sqrt{2}$ for the Normal distribution
- In this course: estimates in S0, with $\delta_1$ also reported

## Three Laws with a Closed-Form Density

- **Normal**: $\alpha = 2$ ($\beta$ plays no role)
  - $\varphi(t) = \exp(-\gamma^2 t^2 + i\delta t)$, i.e. $N(\delta, 2\gamma^2)$
- **Cauchy**: $\alpha = 1$, $\beta = 0$
  - $f(x) = \dfrac{\gamma}{\pi\left(\gamma^2 + (x-\delta)^2\right)}$; $\varphi(t) = \exp(-\gamma|t| + i\delta t)$
- **Lévy**: $\alpha = 1/2$, $\beta = 1$ (S1); totally skewed to the right
  - $f(x) = \sqrt{\dfrac{\gamma}{2\pi}}\,(x-\delta)^{-3/2} \exp\left(-\dfrac{\gamma}{2(x-\delta)}\right)$ for $x > \delta$
  - the first time a Brownian motion hits a level has a Lévy law
- All other stable densities are computed numerically from $\varphi$ Nolan (1997), Mittnik, Doganoglu & Chenyao (1999)

# Normal, Cauchy and Lévy




- Cauchy: lower peak, much heavier tails than the Normal distribution with variance 1
- Lévy: only positive values, a long right tail; its mean is infinite

SFM_ch3_densities

## Worked Example: S1 vs S0 at $t = 1$

- $X$ with $\alpha = 1.5$, $\beta = 0.3$, $\gamma = 1$, $\delta = 0$; note $\tan \frac{3\pi}{4} = -1$
- **S1**: $\ln \varphi(1) = -1 \cdot [1 - i \cdot 0.3 \cdot (-1)] = -1 - 0.3i$
  - $\varphi(1) = e^{-1}(\cos 0.3 - i \sin 0.3) = 0.351 - 0.109i$; $|\varphi(1)| = e^{-1} = 0.368$
- **S0**: the factor $|t|^{1-\alpha} - 1$ is 0 at $t = 1$, so $\varphi(1) = e^{-1} = 0.368$, a real number
  - same modulus: the two laws differ only by a shift
  - shift $= \beta\gamma \tan \frac{\pi\alpha}{2} = -0.3$: the S1 law with $\delta_1 = 0$ is the S0 law with $\delta_0 = -0.3$

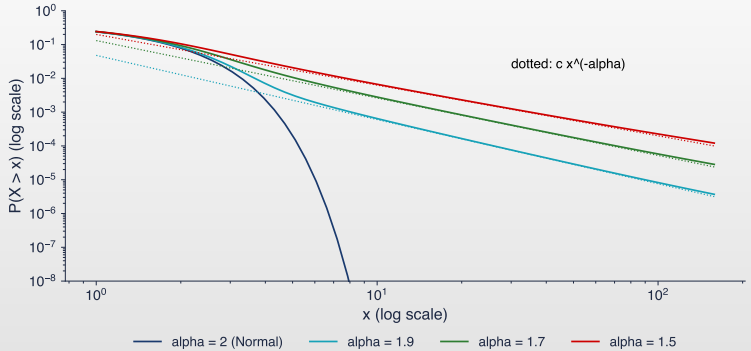
## Recap: Parameters and the Characteristic Function

- Stable laws are defined by $\varphi(t)$; the density is computed numerically
- $\alpha$ tails, $\beta$ skewness, $\gamma$ scale, $\delta$ location
- S0 vs S1: only $\delta$ differs, $\delta_0 = \delta_1 + \beta\gamma \tan \frac{\pi\alpha}{2}$; scipy uses S1 by default
- Closed forms: Normal ($\alpha = 2$, variance $2\gamma^2$), Cauchy, Lévy

## Power-Law Tails

- For $\alpha < 2$, as $x \to \infty$ Nolan (2020):
  - $P(X > x) \sim \gamma^\alpha c_\alpha (1+\beta)\, x^{-\alpha}$ and $P(X < -x) \sim \gamma^\alpha c_\alpha (1-\beta)\, x^{-\alpha}$
  - $c_\alpha = \Gamma(\alpha)\sin(\pi\alpha/2)/\pi$, with $\Gamma$ the gamma function; $f \sim g$ means $f/g \to 1$
- On log-log axes the tail is a straight line with slope $-\alpha$
  - doubling the threshold divides the tail probability by $2^\alpha$: $\times 0.31$ for $\alpha = 1.7$
- The Normal tail falls like $e^{-x^2/(4\gamma^2)}$: doubling the threshold makes it vanish
- $\beta = \pm 1$ with $\alpha < 1$: one tail is missing (the law lives on a half-line)

# Tails on Log-Log Axes



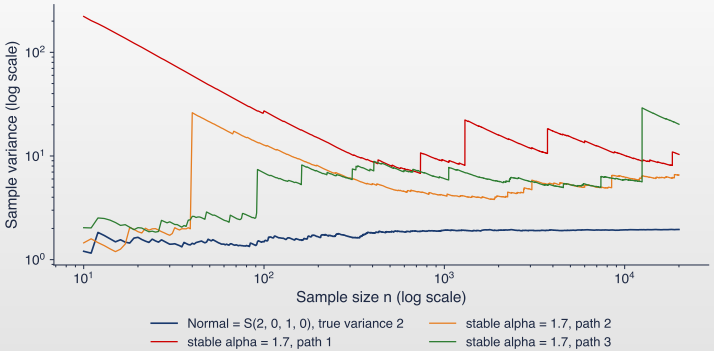

- Solid: $P(X > x)$ of $S(\alpha, 0, 1, 0)$; dotted: the power law $c_\alpha x^{-\alpha}$, reached already near $x = 5$
- $P(X > 10)$: 0.279% for $\alpha = 1.7$, 0.667% for $\alpha = 1.5$, about $8 \times 10^{-11}$% for the Normal distribution

SFM_ch3_tails_moments

## Which Moments Exist?

- For $0 < \alpha < 2$: $E|X|^p < \infty$ if and only if $p < \alpha$
  - $E|X|^p$: the absolute moment of order $p > 0$; $p = 1$ gives the mean, $p = 2$ the variance
  - reason: $|x|^p$ times a density of order $|x|^{-\alpha-1}$ is integrable only if $p < \alpha$
- Consequences
  - $\alpha < 2$: **infinite variance**; skewness and kurtosis are not defined
  - $1 < \alpha < 2$: the mean exists; in S1 it equals $\delta_1$
  - $\alpha \leq 1$: not even the mean exists (Cauchy, Lévy)
  - $\alpha = 2$: all moments exist
- **What do you think?** Mandelbrot found $\alpha \approx 1.7$ for cotton. Does the mean daily change exist?
- **Answer**
  - yes, because $1 < 1.7$; but the variance does not, because $2 > 1.7$

# The Sample Variance Does Not Settle




- Normal (blue): the sample variance converges to the true value 2
- Stable $\alpha = 1.7$: each large draw makes the variance jump; after 20 000 draws the three paths end at 10.4, 6.6, 20.2
- Mandelbrot saw exactly this in cotton prices

SFM_ch3_tails_moments

# Tail Probabilities: Stable vs Normal

| $k$ | $P(\lvert X\rvert > k)$, Normal % | $P(\lvert X\rvert > k)$, $\alpha = 1.7$ % | ratio |
|---|---|---|---|
| 2 | 15.73 | 18.59 | 1.2 |
| 3 | 3.39 | 7.25 | 2.1 |
| 5 | 0.041 | 2.13 | 52.4 |
| 10 | $1.5 \times 10^{-10}$ | 0.56 | $3.6 \times 10^{9}$ |

- Both laws with $\beta = 0$, $\gamma = 1$, $\delta = 0$: the Normal law is $N(0, 2)$
- Near the centre the two laws are close; far in the tails they differ by orders of magnitude
- A risk model is judged in the tails, where the choice of law matters most

SFM_ch3_tails_moments

## Infinite Variance and Risk Measures

- Tools that need a finite variance
  - volatility, the Sharpe ratio and its standard error (Chapter 1); mean–variance portfolios; confidence intervals based on the CLT
  - with $\alpha < 2$ these numbers exist in every sample but estimate nothing
- Tools that survive
  - quantiles always exist: $\mathrm{VaR}_{1\%} = -q_{1\%}$, the loss exceeded with probability 1%
  - $\mathrm{ES}_{2.5\%}$ (expected shortfall, the mean loss beyond the VaR) exists only if $\alpha > 1$
  - the scale $\gamma$ and the interquartile range replace the standard deviation
- VaR and ES are developed in Chapter 10

## Recap: Tails and Moments

- Tails: $P(|X| > x) \propto x^{-\alpha}$; a straight line on log-log axes
- Moments: $E|X|^p < \infty$ iff $p < \alpha$; $\alpha < 2$ means infinite variance
- The sample variance of a stable sample jumps and does not settle
- Quantiles, VaR and the scale $\gamma$ remain meaningful

## The Chambers–Mallows–Stuck Method (Symmetric Case)

- No closed-form density, yet stable variables are easy to simulate Chambers, Mallows & Stuck (1976)
- **CMS** (Chambers–Mallows–Stuck) for $S(\alpha, 0, 1, 0)$:
  - draw $V \sim U(-\pi/2, \pi/2)$, a uniform angle, and $W \sim \mathrm{Exp}(1)$, an exponential variable with mean 1, independent
  - $X = \dfrac{\sin(\alpha V)}{(\cos V)^{1/\alpha}} \left( \dfrac{\cos((1-\alpha)V)}{W} \right)^{(1-\alpha)/\alpha}$
  - then $\gamma X + \delta \sim S(\alpha, 0, \gamma, \delta)$
- Checks on special cases
  - $\alpha = 1$: $X = \tan V$, a Cauchy variable
  - $\alpha = 2$: $X = 2 \sin V \sqrt{W}$, a $N(0, 2)$ variable

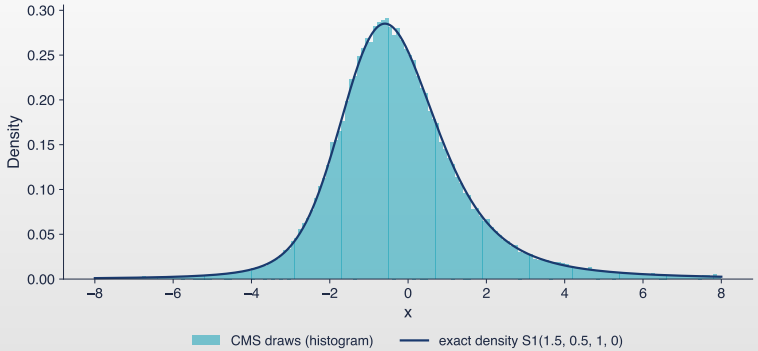
## The General Case ($\beta \neq 0$)

- For $\alpha \neq 1$, S1 parameterisation Weron (1996):
  - $B = \dfrac{\arctan(\beta \tan \frac{\pi\alpha}{2})}{\alpha}, \quad S = \left(1 + \beta^2 \tan^2 \frac{\pi\alpha}{2}\right)^{1/(2\alpha)}$
  - $X = S \dfrac{\sin(\alpha(V+B))}{(\cos V)^{1/\alpha}} \left(\dfrac{\cos(V - \alpha(V+B))}{W}\right)^{(1-\alpha)/\alpha} \sim S(\alpha, \beta, 1, 0; 1)$
- $B$, $S$: auxiliary constants that depend only on $\alpha$ and $\beta$; for $\beta = 0$, $B = 0$ and $S = 1$
- From the standard draw to any parameters
  - S1: $\gamma X + \delta_1$; S0: $\gamma X + \delta_0 - \beta\gamma \tan \frac{\pi\alpha}{2}$
- Weron (1996) corrected a formula of the original paper for $\beta \neq 0$; `scipy.stats.levy_stable.rvs` uses the same method

## Worked Example: One Draw Step by Step

- $\alpha = 1.5$, $\beta = 0$; a uniform draw $U = 0.8$ and an exponential draw $W = 1.2$
- Steps
  - $V = \pi(U - 1/2) = 0.942$: turns $U$, uniform on $(0, 1)$, into a uniform angle on $(-\pi/2, \pi/2)$
  - $\sin(\alpha V) = 0.988$; $(\cos V)^{1/\alpha} = 0.702$
  - $\big(\cos((1-\alpha)V)/W\big)^{(1-\alpha)/\alpha} = 1.104$
  - $X = 0.988/0.702 \times 1.104 = 1.554$
- Repeat with new $(U, W)$ pairs to obtain a sample of any size

# CMS Draws against the Exact Density




- 100 000 CMS draws of $S(1.5, 0.5, 1, 0; 1)$: the histogram follows the density computed from $\varphi$
- Quantiles 1% and 99%: sample $-5.47$ and 9.52, exact (scipy `levy_stable`) $-5.39$ and 9.79
- Two-sample KS (Kolmogorov–Smirnov) test against `levy_stable.rvs`: statistic 0.0044, p-value 0.30, no difference

SFM_ch3_sim_stable

## Recap: Simulation

- CMS: a uniform angle $V$ and an exponential $W$ give one stable draw
- Special cases check the formula: $\tan V$ (Cauchy), $2 \sin V \sqrt{W}$ (Normal)
- Skewed case: the formula of Weron (1996); shift by $\beta\gamma \tan \frac{\pi\alpha}{2}$ for S0

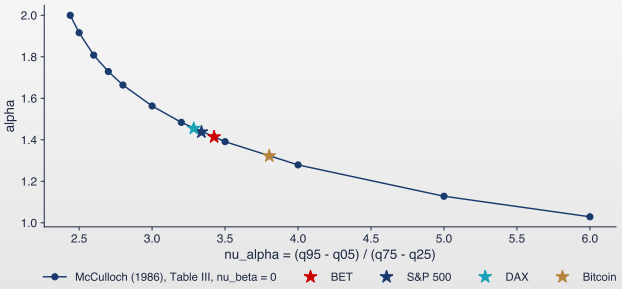
## Three Families of Estimators

- **Quantile methods**: match sample quantiles to those of the stable law
  - Fama & Roll (1971) for symmetric laws; McCulloch (1986) for all four parameters
- **Characteristic-function methods**: regress the empirical $\ln(-\ln|\hat{\varphi}(t)|^2)$ on $\ln|t|$
  - $\hat{\varphi}(t) = \frac{1}{n}\sum_{j=1}^{n} e^{itr_j}$: the empirical characteristic function; $|\varphi(t)|^2 = e^{-2\gamma^\alpha |t|^\alpha}$, so the log-log relation is linear in $\ln|t|$
  - Koutrouvelis (1980); the slope estimates $\alpha$
- **Maximum likelihood** (ML): the most precise, once the density can be computed
  - Nolan (2001), Mittnik et al. (1999)
- A SAS implementation of these estimators is described in Pele (2014)

## McCulloch's Quantile Method

- Two ratios of sample quantiles $\hat{q}_p$, free of location and scale McCulloch (1986):
  - $\nu_\alpha = \dfrac{\hat{q}_{0.95} - \hat{q}_{0.05}}{\hat{q}_{0.75} - \hat{q}_{0.25}}$ (tail width relative to the body)
  - $\nu_\beta = \dfrac{\hat{q}_{0.95} + \hat{q}_{0.05} - 2\hat{q}_{0.5}}{\hat{q}_{0.95} - \hat{q}_{0.05}}$ (asymmetry)
- $\hat{q}_p$: the sample quantile of order $p$, the value below which a share $p$ of the returns lies
- Tables III and IV of the paper map $(\nu_\alpha, \nu_\beta)$ to $(\hat{\alpha}, \hat{\beta})$
  - Normal distribution: $\nu_\alpha = 2.439$; larger $\nu_\alpha$ means smaller $\alpha$
- Then $\hat{\gamma} = (\hat{q}_{0.75} - \hat{q}_{0.25})/\phi_3(\hat{\alpha}, \hat{\beta})$ (Table V) and $\hat{\delta}_0$ from Table VII; $\phi_3$: the interquartile range of $S(\alpha, \beta, 1, 0)$
- Fast, consistent, valid for $0.6 \leq \alpha \leq 2$; less precise than ML; a good starting point for ML

# McCulloch's Map from $\nu_\alpha$ to $\alpha$




- Line: Table III for a symmetric sample ($\nu_\beta = 0$)
- Stars: daily log returns since 2000 (Bitcoin since 2014)
- $\nu_\alpha$: BET 3.425, S&P 500 3.338, DAX 3.285, Bitcoin 3.803
- All well above 2.439: tails much wider than the Normal distribution allows

SFM_ch3_estimation

## Worked Example: McCulloch on the BET

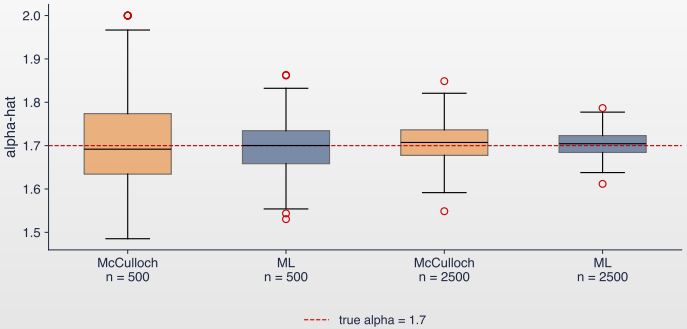
- BET daily log returns (%), 2000–2026, $n = 6\,670$
- Step 1: quantiles
  - $\hat{q}_{0.05} = -1.9345$, $\hat{q}_{0.25} = -0.5025$, $\hat{q}_{0.5} = 0.0771$, $\hat{q}_{0.75} = 0.6649$, $\hat{q}_{0.95} = 2.0641$
- Step 2: ratios
  - $\nu_\alpha = 3.9986/1.1674 = 3.425$; $\nu_\beta = -0.006$, practically symmetric
- Step 3: Table III, column $\nu_\beta = 0$: $\nu_\alpha = 3.2 \mapsto 1.484$, $\nu_\alpha = 3.5 \mapsto 1.391$
  - linear interpolation: $\hat{\alpha} \approx 1.414$
- Step 4: $\hat{\gamma} = 0.60$ (% a day), $\hat{\beta} = -0.011$, $\hat{\delta}_0 = 0.079$

SFM_ch3_estimation

## Maximum Likelihood

- Log-likelihood of a sample $r_1, \ldots, r_n$: $\ell(\theta) = \sum_{t=1}^{n} \ln f(r_t; \theta)$, $\theta = (\alpha, \beta, \gamma, \delta_0)$
  - $\theta$: the vector of the four parameters; $f(r_t; \theta)$: the stable density at $r_t$; $\hat{\theta}$ maximises $\ell$ numerically, starting from the McCulloch estimate
  - standard errors from the curvature of $\ell$: the inverse of the Hessian, the matrix of second derivatives of $\ell$
- The density $f$ is computed from $\varphi$
  - $f(x) = \frac{1}{2\pi} \int e^{-itx} \varphi(t)\, dt$, evaluated on a grid with the FFT (fast Fourier transform) Mittnik, Doganoglu & Chenyao (1999)
  - or by direct integration Nolan (1997); `levy_stable.fit` does this and needs minutes for a few thousand returns
- ML in S0 is regular for $\alpha \in (0, 2)$: standard errors shrink like $1/\sqrt{n}$ Nolan (2001)

# McCulloch vs ML in a Monte Carlo Study




- 200 samples of $S(1.7, 0, 1, 0)$ for each size; both estimators are centred on 1.7
- Standard deviation of $\hat{\alpha}$: $n = 500$: McCulloch 0.1065, ML 0.0611; $n = 2500$: 0.0462 vs 0.0298
- ML is more precise for the same data: McCulloch's standard deviation is 1.74 times that of ML at $n = 500$, 1.55 times at $n = 2500$

SFM_ch3_estimation

## Recap: Estimation

- McCulloch: five quantiles, two ratios, four tables; fast and robust
- ML: numerical density from $\varphi$, numerical maximisation; the most precise
- Report the parameterisation (S0 or S1) and the units of $\gamma$ and $\delta$

## Stable Fits of Daily Log Returns

| | $n$ | $\hat{\alpha}$ McC. | $\hat{\alpha}$ ML (SE) | $\hat{\beta}$ ML (SE) | $\hat{\gamma}$ | $\hat{\delta}_0$ | $\hat{\delta}_1$ |
|---|---|---|---|---|---|---|---|
| BET | 6 670 | 1.414 | 1.486 (0.020) | −0.016 (0.034) | 0.62 | 0.088 | 0.078 |
| S&P 500 | 6 714 | 1.438 | 1.540 (0.021) | −0.181 (0.036) | 0.58 | 0.093 | 0.000 |
| DAX | 6 786 | 1.455 | 1.606 (0.021) | −0.179 (0.040) | 0.74 | 0.090 | −0.004 |
| Bitcoin | 4 384 | 1.322 | 1.410 (0.026) | 0.019 (0.036) | 1.54 | 0.132 | 0.171 |

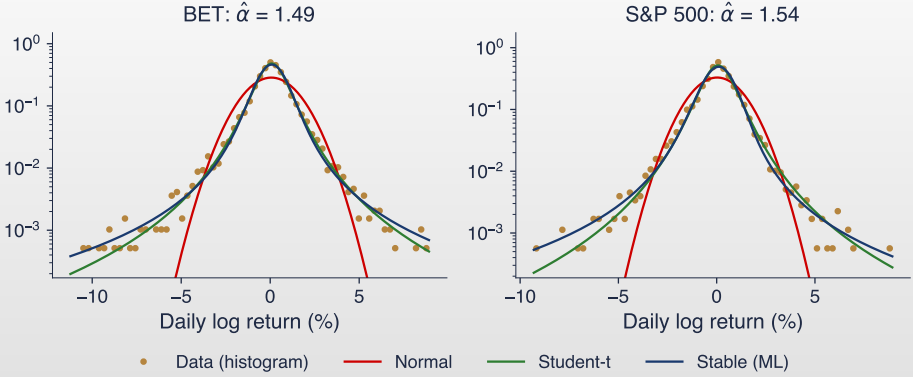
- Daily log returns in %; indices 2000–2026, Bitcoin 2014–2026 (7 days a week); data: EODHD
- McC.: McCulloch; ML in S0 with standard errors; $\hat{\gamma}$, $\hat{\delta}_0$, $\hat{\delta}_1$ in % a day

SFM_ch3_fit_returns

## Reading the Estimates

- $\hat{\alpha}$ well below 2 for all four series
  - S&P 500: $(2 - 1.540)/0.021 \approx 22$ standard errors from the Normal case
  - the heaviest tails: Bitcoin ($\hat{\alpha} = 1.410$)
- $\hat{\beta} < 0$ for the S&P 500 and the DAX: the left tail is heavier (crashes)
  - BET and Bitcoin: $\hat{\beta}$ within two standard errors of 0
- $\hat{\gamma}$: Bitcoin 1.54% vs S&P 500 0.58% a day; compare the sample standard deviations 3.49% and 1.21%
- McCulloch gives lower $\hat{\alpha}$ than ML for every series: the two methods weigh the tails differently

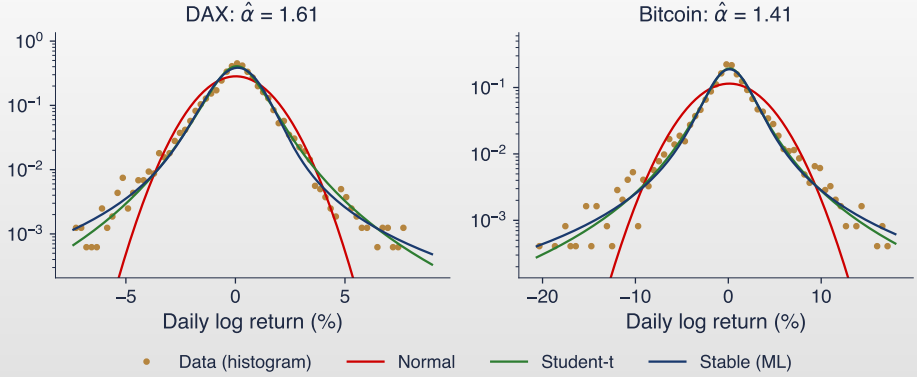
# Fitted Densities on a Log Scale (1/2)




- Normal (red): far too few large returns; Student-t (green) and stable (blue) both follow the body

SFM_ch3_fit_returns

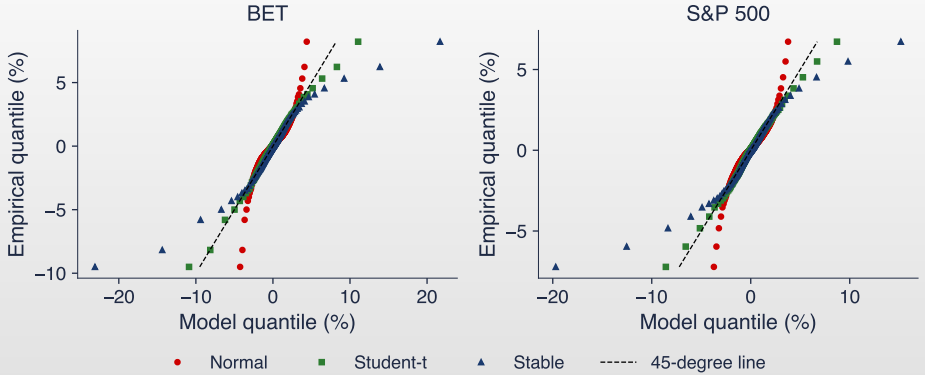
# Fitted Densities on a Log Scale (2/2)




- Far tails: the stable density stays above the data points; the Student-t is closer

SFM_ch3_fit_returns

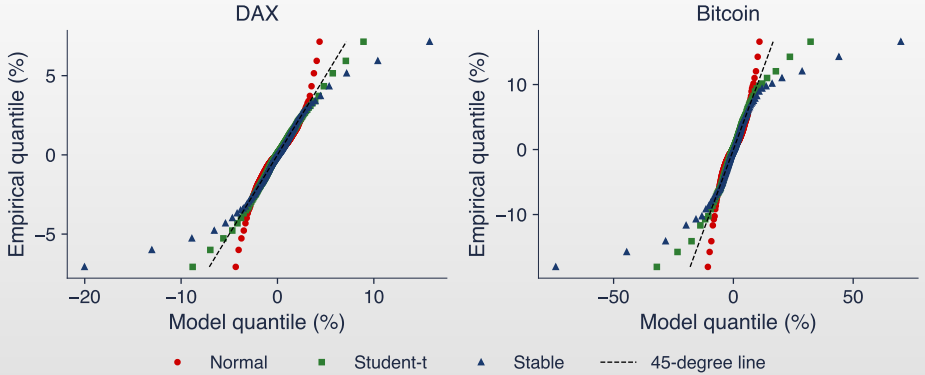
# QQ Plots against the Three Models (1/2)




- A QQ (quantile–quantile) plot puts the empirical quantiles against the model quantiles; a good model lies on the 45-degree line

SFM_ch3_fit_returns

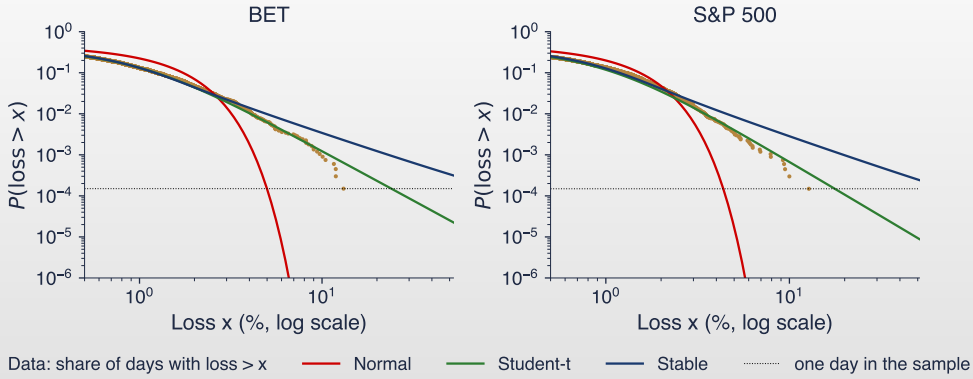
# QQ Plots against the Three Models (2/2)




- Normal: too-short tails (steep ends); stable: too-long tails (flat ends, model quantiles far beyond the data); Student-t: the closest

SFM_ch3_fit_returns

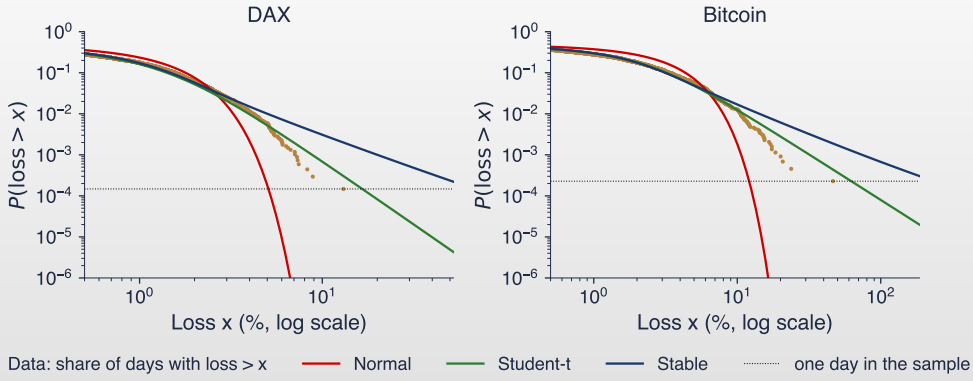
# The Left Tail on Log-Log Axes (1/2)




- Points: share of days with a loss above $x$; lines: the same probability under each fitted model

SFM_ch3_fit_returns

# The Left Tail on Log-Log Axes (2/2)




- The empirical tail bends down faster than the stable line: the tail index of the data is larger than $\hat{\alpha}$

SFM_ch3_fit_returns

## Counting Large Losses

| | loss above | observed | Normal | Student-t | stable |
|---|---|---|---|---|---|
| S&P 500 | 5% | 23 | 0.1 | 27.0 | 57.3 |
| S&P 500 | 10% | 1 | $\approx 0$ | 4.5 | 19.1 |
| S&P 500 | 15% | 0 | $\approx 0$ | 1.5 | 10.2 |
| BET | 10% | 6 | $\approx 0$ | 8.2 | 22.8 |
| Bitcoin | 10% | 56 | 8.1 | 54.2 | 73.9 |
| Bitcoin | 20% | 4 | $\approx 0$ | 12.3 | 26.9 |

- Expected number of days $= n \times P(r < -x)$ under each fitted model
- Normal: almost no large losses; stable: too many, and the gap grows with the threshold; Student-t: closest to the counts
- S&P 500: 1 day below $-10\%$ in 6 714 days; the stable fit expects 19.1

SFM_ch3_fit_returns

## VaR 1% and VaR 0.1% under the Three Models

| | VaR 1% (%) | | | | VaR 0.1% (%) | | | |
|---|---|---|---|---|---|---|---|---|
| | data | Normal | t | stable | data | Normal | t | stable |
| BET | 4.1 | 3.2 | 4.1 | 4.9 | 9.5 | 4.3 | 10.9 | 23.1 |
| S&P 500 | 3.5 | 2.8 | 3.5 | 4.5 | 7.2 | 3.7 | 8.6 | 19.7 |
| DAX | 4.2 | 3.3 | 4.0 | 5.0 | 7.1 | 4.3 | 8.8 | 20.0 |
| Bitcoin | 10.5 | 8.0 | 11.1 | 14.3 | 18.1 | 10.7 | 31.9 | 74.2 |

- VaR at level $p$: $\mathrm{VaR}_p = -q_p$, the daily loss exceeded with probability $p$ (Chapter 10); here $p$, because $\alpha$ is the stable index
- VaR 1%: the Normal model is too low, the stable model too high
- VaR 0.1%: the stable model gives roughly two to four times the empirical quantile

SFM_ch3_fit_returns

## Which Model Fits Best?

| | AIC Normal | AIC Student-t | AIC stable | $\hat{\nu}$ (Student-t) |
|---|---|---|---|---|
| BET | 23 407 | 20 996 | 21 088 | 2.43 |
| S&P 500 | 21 650 | 19 702 | 19 794 | 2.67 |
| DAX | 23 901 | 22 595 | 22 697 | 3.09 |
| Bitcoin | 23 391 | 22 056 | 22 159 | 2.23 |

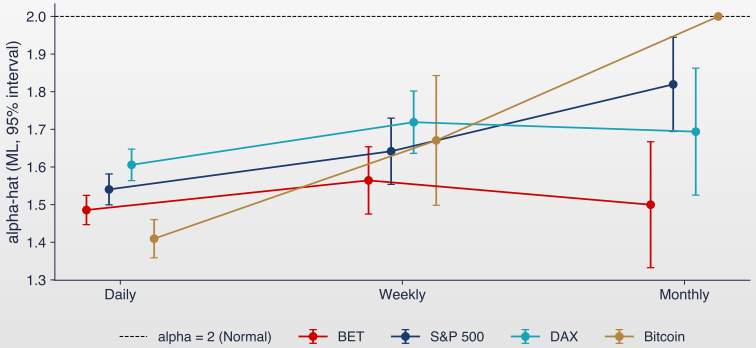
- AIC (Akaike information criterion) $= 2k - 2\ell(\hat{\theta})$, with $k$ the number of parameters; lower is better
- Both heavy-tailed models beat the Normal distribution: Student-t by 1 306 (DAX) to 2 411 (BET) AIC points, stable by 1 204 to 2 319
- Student-t wins for every series, with fewer parameters; S&P 500: by 92 points
- $\hat{\nu}$ between 2 and 3.1: finite variance, but few finite moments beyond it

SFM_ch3_fit_returns

## Recap: Fitting Real Returns

- All four series: $\hat{\alpha}$ between 1.410 and 1.606, far from 2
- The stable law fits the body much better than the Normal distribution
- It overstates the far tails: too many extreme losses, VaR 0.1% too high
- By AIC, the Student-t distribution fits better

# Test 1: Does $\alpha$ Stay the Same When Returns Are Summed?



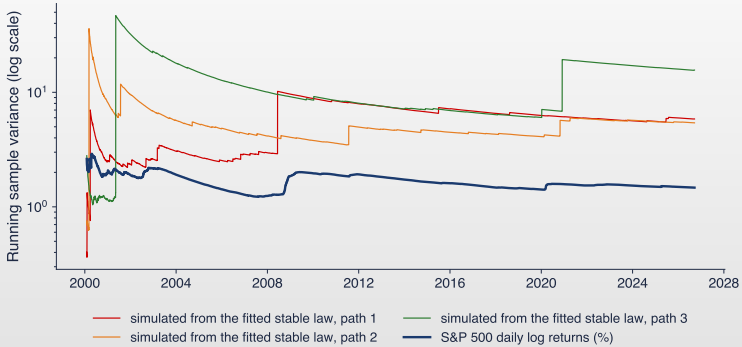

- If daily returns were i.i.d. stable, weekly and monthly sums would have the **same** $\alpha$ (stability)
- ML estimates: S&P 500 1.54 $\rightarrow$ 1.64 $\rightarrow$ 1.82; Bitcoin 1.41 $\rightarrow$ 1.67 $\rightarrow$ 2.00 (at the bound 2)

SFM_ch3_aggregation

## Reading the Aggregation Test

- $\hat{\alpha}$ rises with the horizon for the S&P 500, DAX and Bitcoin
  - DAX: $1.61 \to 1.72 \to 1.69$; the daily and monthly intervals do not overlap for the S&P 500
- BET: $1.49 \to 1.56 \to 1.50$, no clear rise
  - monthly samples are short (321 months): wide intervals
- The same pattern in the classic studies: Officer (1972), Akgiray & Booth (1988)
- A rising $\alpha$ is what the GCLT predicts for sums of finite-variance returns that converge slowly to the Normal distribution

# Test 2: Does the Sample Variance Settle?




- S&P 500 (blue): after the jumps of 2008 and 2020 the running variance settles near 1.47
- Paths simulated from the fitted stable law, same length: final variances between 5.4 and 15.6, far above the data

SFM_ch3_aggregation

## Test 3: How Heavy Is the Far Tail?

- The stable $\alpha$ is driven by the whole distribution, mostly by the body
  - the tail index of the far tail alone is estimated with the Hill estimator Hill (1975) (Chapter 5)
- For stock indices the far tail decays roughly like $x^{-3}$, the "inverse cubic law" Gopikrishnan et al. (1999)
  - a tail index near 3 means finite variance, and no stable law has it
- Our counts agree: the stable fits expect many more losses beyond 10% than observed
- Lau, Lau & Wingender (1990): further evidence against the stable model from the behaviour of sample moments

## Why Do Daily Returns Look Stable?

- **Volatility clustering** (Chapters 2 and 9): calm and turbulent periods alternate
  - a mixture of Normal laws with changing variance has heavy tails and a finite variance Blattberg & Gonedes (1974)
  - fitted to one stable law, the mixture gives $\hat{\alpha} < 2$
- **Truncated and tempered stable laws**: stable at moderate scales, lighter tails far out
  - Mantegna & Stanley (1995) fitted a truncated Lévy flight to the S&P 500
  - Grabchak & Samorodnitsky (2010): such laws explain why returns look stable at short horizons and Normal at long ones

## Verdict

- **Mandelbrot was right** about the facts
  - daily returns have heavy tails; the Normal distribution fails badly
- **The infinite variance is not supported**
  - $\hat{\alpha}$ rises with aggregation; the sample variance settles; the far tail is lighter than any stable tail
- Where stable laws remain useful
  - a benchmark for heavy tails; simulation; the scaling idea; data with truly infinite variance (some insurance losses, some crypto tokens)
  - the GCLT: a reminder that $\sqrt{n}$-rules need a finite variance
- Next: Chapter 5 estimates the tail index directly with extreme value theory

## Recap: The Critique

- Stability predicts the same $\alpha$ at every horizon: the data show a rising $\alpha$
- The sample variance of real returns settles; that of stable samples does not
- Volatility clustering and tempered tails explain the apparent $\alpha < 2$
- Heavy tails: yes; infinite variance: no

## An Open Question

- **Have the tails of Bitcoin become lighter as the market matured?**
  - more participants, futures (since 2017) and spot ETFs (exchange-traded funds, since 2024)
  - if the tails got lighter, risk models fitted on old data overstate today's risk
- Why it is open: $\hat{\alpha}$ moves with volatility regimes, and yearly samples are short
- Full sample: $\hat{\alpha} = 1.410$ (SE 0.026), the lowest of our four series
- AI tools can speed up such a study; they do not replace checking it Wang et al. (2023)

## How AI Could Help

- **Literature**: list papers on tail estimates for crypto assets and summarise their methods
- **Code**: draft a rolling-window McCulloch estimator with bootstrap intervals
- **Robustness**: propose other windows, ML instead of quantiles, Student-t $\nu$ instead of $\alpha$
- **Explanation**: a first draft of the interpretation of a table of yearly estimates
- Example prompt
  - `Write a Python function that estimates the stable alpha of daily log returns on rolling 365-day windows with McCulloch (1986), and adds 95% bootstrap intervals.`

## What to Check

- Parameterisation: S0 or S1; scipy uses S1 by default; the location differs by $\beta\gamma \tan \frac{\pi\alpha}{2}$
- Units: returns in % or in decimals; $\gamma$ and $\delta$ change, $\alpha$ and $\beta$ do not
- Overlapping windows are not independent: do not read the rolling estimates as separate tests
- Numbers: recompute one window with a second method (ML or the quantile tables)
- References: every cited paper must exist; check the DOI

## Project Seed

- **Question**: is the tail of Bitcoin returns lighter now than in 2014–2018?
  - data: Bitcoin daily closes from the course data (EODHD), 2014–2026
- Steps
  - estimate $\alpha$ by McCulloch and by ML for each calendar year, with bootstrap intervals
  - repeat on returns divided by a rolling volatility: does the trend survive?
  - compare with the Student-t $\hat{\nu}$ of each year
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

# Key Formulas

| Quantity | Formula |
|---|---|
| Stability | $aX_1 + bX_2 \stackrel{d}{=} cX + d, \quad c^\alpha = a^\alpha + b^\alpha$ |
| Sum of $n$ copies | $X_1 + \cdots + X_n \stackrel{d}{=} n^{1/\alpha} X + d_n$ |
| Characteristic function (S1) | $\exp\{-\gamma^\alpha \lvert t\rvert^\alpha [1 - i\beta \operatorname{sign}(t) \tan \frac{\pi\alpha}{2}] + i\delta_1 t\}$ |
| S0 vs S1 | $\delta_0 = \delta_1 + \beta\gamma \tan \frac{\pi\alpha}{2}$ |
| Normal case | $S(2, \beta, \gamma, \delta) = N(\delta, 2\gamma^2)$ |
| Tails | $P(X > x) \sim \gamma^\alpha c_\alpha (1+\beta) x^{-\alpha}$ |
| Moments | $E\lvert X\rvert^p < \infty \iff p < \alpha \quad (\alpha < 2)$ |
| CMS ($\beta = 0$) | $\frac{\sin(\alpha V)}{(\cos V)^{1/\alpha}} \left(\frac{\cos((1-\alpha)V)}{W}\right)^{(1-\alpha)/\alpha}$ |
| McCulloch | $\nu_\alpha = \frac{q_{0.95} - q_{0.05}}{q_{0.75} - q_{0.25}}, \quad \nu_\beta = \frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$ |

## Check Yourself

- **Question**: daily returns are i.i.d. $S(1.5, 0, \gamma, 0)$; by what factor does the scale grow from 1 day to 20 days?
  - **Answer**: $20^{1/1.5} = 7.37$, i.e. 1.65 times the square-root-of-time factor $\sqrt{20}$
- **Question**: a fitted law has $\alpha = 1.2$; do the mean and the variance exist?
  - **Answer**: the mean exists ($1 < 1.2$), the variance does not ($2 > 1.2$)
- **Question**: scipy reports $\delta = 0.05$ with $\alpha = 1.7$, $\beta = -0.2$, $\gamma = 0.6$; what is $\delta_0$?
  - **Answer**: scipy uses S1, so $\delta_0 = 0.05 + (-0.2)(0.6)\tan(0.85\pi) = 0.05 + 0.061 = 0.111$
- Next: Chapter 4, probability for finance

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